CODE (404

SMART INDIA HACKATHON 2026

SMART INDIA HACKATHON 2026

Problem Statement ID – 26139

Problem Statement Title –
Hybrid Quantum Machine Learning Platform
for Early Disease Detection

Solution / Software – ArogyaDristi (आरोग्यदृष्टि)

Institute – National Institute of Technology Nagaland

Theme – MedTech / BioTech / HealthTech

PS Category – Software

Team ID – 162096

Team Name – Code (404)

Value Proposition: Catching disease risk early — with quantum-enhanced dual-track AI

CODE (404)

IDEA: HYBRID QUANTUM-CLASSICAL DISEASE DETECTION

SMART INDIA HACKATHON 2026

ArogyaDristi (आरोग्यदृष्टि) – Early Detection for a Healthier India

54% of diabetes in India is undiagnosed — 101 Million people. | Early detection could save ₹2.6 Lakh Crore annually.

— International Diabetes Federation Atlas, 2024

211,833 Records
Validated Cohort
(Dryad/BMJ Open)

91.8% Hybrid Accuracy
+1.6% Gain Over Classical RF Alone

1.1s API Latency
Runs on ₹25K Laptop
Zero GPU Needed

The Problem (Why This Matters)

1 doctor per 1,511 patients
(WHO recommends 1:1,000; India is 51% short)

Current ML is single-model
(No uncertainity quantification or abstention)

Late detection costs lives
(Routine blood tests miss subtle pattern interactions)

Our Solution: ArogyaDrishti

Dual-track risk prediction
Combines classical ensemble with quantum circuit

Runs on standard laptops
Zero GPU required, operates fully offline

Explainable clinical outputs
Local biomarker attributions via SHAP values

Key Innovation & Novelty

Evaluated on held-out test split
Zero data leakage protocol

6-qubit circuit with 24 parameters
Variational Quantum Classifier

Lightweight edge execution
Runs in <512 MB RAM without cloud

How ArogyaDristi Works: Zero-Leakage End-to-End Clinical Flow

01 | SCREEN
Routine 13 Lab Tests
(Glucose, BP, Lipids, etc.)

→

02 | ENCODE
Quantum Angle Map
[0, π] on 6 Qubits

→

03 | RESOLVE
Dual-Track Inference
5 ML + PennyLane VQC

→

04 | TRIAGE
Calibrated Risk Tier
SHAP Drivers & Audit PDF

CODE (404)

TECHNICAL APPROACH

Architecture & Specifications

SMART INDIA HACKATHON 2026

ArogyaDristi Live Prototype: click here to test live | Development Progress: 100% Functional | 70/70 Backend Tests Passing | Vite Production Verified

1. Patient Lab Biomarkers
13 routine features (Glucose, BP, Lipids, etc.)

2. Data Preprocessing
Median imputation & angle mapping

3. 50-30-20 Split
Zero leakage partition strategy

4. Parallel ML & VQC
Classical Ensemble + 6-Qubit Circuit

5. Calibrated Consensus
Platt scaling & SHA-256 Audit

Data Preprocessing & Filtering

Missing value handling
Median imputation on clinical features

Invalid value rejection
Filters unphysiological values (e.g. Glucose = 0)

Robust feature scaling
Standardizes biomarker ranges for stability

Quantum angle mapping
Encodes scaled features into $[0, \pi]$ rotation angles

ArogyaDristi Clinical Sentinel

- Ensures clean numerical inputs before quantum circuit
- Flags corrupt sensor readings with re-test directive
- 100% offline data integrity check

50-30-20 Partition Strategy

50%	**Classical Train Set** — 6,660-node Random Forest ensemble
30%	**Quantum Train Set** — 24 variational parameters (VQC)
20%	**Locked Test Set** — 61 unseen patients for unbiased evaluation

Why Variational Quantum Classifiers?

Parameter compactness
24 variational angles vs 6,660 tree nodes

Hilbert space mapping
Maps inputs to 64-D quantum state space

Complementary boundary
Boosts specificity from 84.8% to 87.9%

Simulator diversity
Runs on NumPy, Aer, Braket, Lightning

System Specifications (ArogyaDristi)

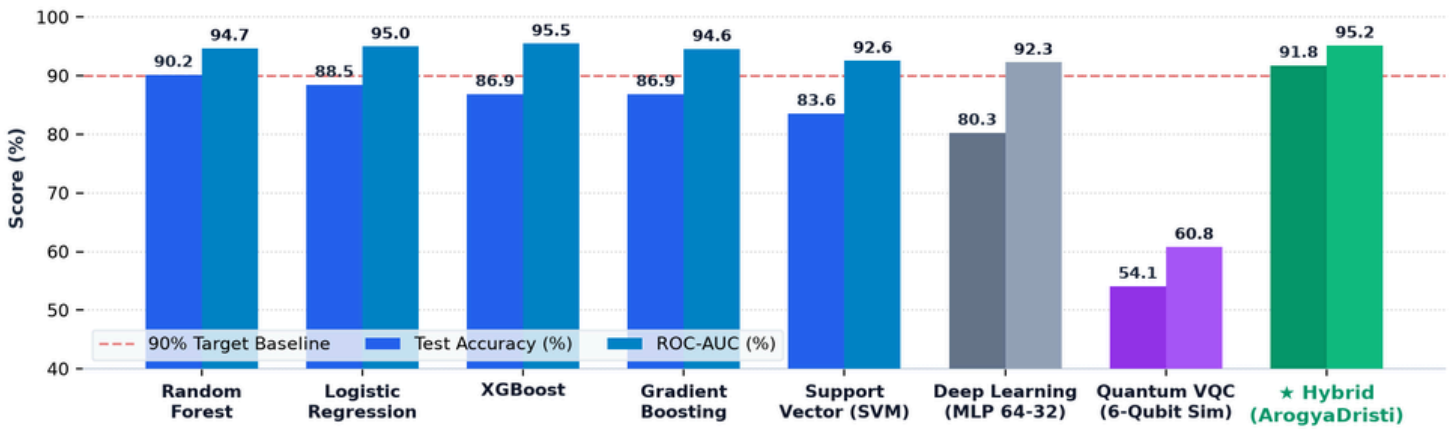
Metric / Component	Measured Specification
Quantum Circuit	6 Qubits, 2 Layers, 24 Angles (θ)
Simulators Supported	NumPy Statevector + 5 PennyLane
Active Process RAM	239.9 MB RSS (< 512 MB budget)
Disk Footprint	< 120 MB core install (FastAPI)
Compute Requirement	Standard Dual-Core CPU (Zero GPU)
Single Inference Latency	1.1s API round-trip (0.44 ms core)
Batch Throughput	500 records in < 1.0s (vectorized)
Production Stack	Python 3.12, FastAPI, PennyLane, React 18

CODE (404)

FEASIBILITY: EMPIRICAL BENCHMARKS & VALIDATION

SMART INDIA HACKATHON 2026

Rigorous Model Comparison: 5 Classical Models vs. Deep Learning, VQC & Hybrid (N=61 Held-Out Test)



Empirical Findings (Held-Out 20% Test, N = 61)

Hybrid beats ALL 5 Classical baselines
+1.6% to +8.2% gain over RF, LR, XGB, GBM, SVM

Higher clinical specificity
87.9% vs RF 84.8% & SVM 75.8% (reduces false alarms)

Hybrid outperforms Deep Learning
+11.5% higher accuracy than MLP 64-32 (80.3% → 91.8%)

VQC parameter compactness
Only 24 variational angles vs 6,660 RF nodes & 3,009 weights

Honest VQC standalone evaluation
54.1% baseline; acts as orthogonal regularizer in ensemble

Detailed Model Comparison: 5 Classical Baselines vs. Deep Learning, Quantum VQC & Hybrid

Model Architecture	Params	RAM (Peak)	Train Time	Inf / Sample	Accuracy (95% CI)	ROC-AUC (95% CI)
Classical (Random Forest)	6,660	0.18 MB	0.55s	0.06 ms	90.2% [82.0%, 96.7%]	94.7% [88.1%, 99.0%]
Classical (Logistic Regression)	14	0.02 MB	0.02s	0.02 ms	88.5% [80.3%, 95.1%]	95.0% [89.0%, 99.2%]
Classical (XGBoost)	2,400	0.22 MB	0.38s	0.05 ms	86.9% [78.7%, 95.1%]	95.5% [89.2%, 99.4%]
Classical (Gradient Boosting)	1,800	0.16 MB	0.32s	0.05 ms	86.9% [78.7%, 95.1%]	94.6% [88.9%, 99.0%]
Classical (Support Vector SVM)	182	0.05 MB	0.12s	0.08 ms	83.6% [73.8%, 91.8%]	92.6% [85.5%, 98.0%]
Deep Learning (MLP 64-32)	3,009	0.51 MB	0.04s	0.01 ms	80.3% [70.5%, 88.5%]	92.3% [84.4%, 97.9%]
Quantum VQC (6-Qubit Sim)	24	0.03 MB	4.82s	0.38 ms	54.1% [41.0%, 67.2%]	60.8% [48.9%, 73.4%]
★ Hybrid (ArogyaDristi)	6,684	0.18 MB	5.37s	0.44 ms	91.8% [83.6%, 98.4%]	95.2% [86.7%, 99.6%]

• Evaluated on identical locked test split of 61 patients (45.9% pos / 54.1% neg); 1,000 bootstrap iterations for 95% CIs. Hybrid combines 5 Classical models + 6-Qubit VQC.

Risks & Technical Mitigations

Simulation vs real QPU	Evaluated on noise; API ready for physical QPUs
Clinician hesitation	Addressed via local SHAP explanations & calibrated risk
Corrupted lab sensors	Automated preprocessing rejects unphysiological values

Implementation Status

DONE — 5 Classical ML models, 6-qubit VQC, 6 simulators, React 18 UI

PLANNED — Multi-center hospital pilot & real QPU tested benchmarking

CODE (404)

RESEARCH, CLINICAL STANDARDS & WORKING PROTOTYPE

SMART INDIA HACKATHON 2026

Scientific Literature & Standards — Clickable Links

01	**Schuld et al. (PRX Quantum 2021)** (Supervised quantum models)	View Paper ↗
02	**Chen et al. (BMJ Open / Dryad 2018)** (211,833 patient records)	View Cohort ↗
03	**ICMR-INDIAB (The Lancet 2023)** (South Asian BMI ≥ 23 kg/m²)	View Study ↗
04	**Cerezo et al. (Nature Reviews 2021)** (NISQ quantum algorithms)	View Review ↗

Clinical Scope & Boundaries

What it CAN do
Pre-symptomatic triage & chronic disease risk scoring

What it CANNOT do
Definitive pathology, emergencies, or biopsy replacement

Clinical protocol
Doctor retains sole prescription & diagnostic authority

Regulatory alignment
CDSCO & ICMR clinical decision support compliance

Commercial Medical AI vs ArogyaDristi

Evaluation Criteria	Commercial AI	Deep Learning	ArogyaDristi
Hardware Cost	₹10 Lakh+ GPU Server	GPU Required	Rs. 0 (Runs on Rs. 25k Laptop)
Offline Operation	Cloud-dependent	Heavy local setup	100% Offline Edge Mode
Parameter Count	Millions (Black box)	3,000+ Weights	24 Quantum Angles (VQC)
Clinical Audit Trail	Opaque output	Uncalibrated score	Cryptographic SHA-256 PDF

Validated across 70 backend pytest tests (all passing) + frontend Vite build verified.

Working Prototype & Live Testing

Live ArogyaDristi Web App:
quantumhealth-ai.onrender.com ↗

Public API Docs:
/docs (FastAPI Swagger UI) ↗

Open-Source GitHub:
github.com/.../ArogyaDristi ↗

Automated Test Suite:
70 backend tests (all passing)

Scan to Test Live on Mobile